Branch Coverage Report

Generated 2026-11-12T16:45:25.898236+00:00. Application source: 6b5464298c949806ebba0b7bd1d903c962b5255d.

Environment: Linux-7.2.14-200.fc44.x86_64-x86_64-with-glibc2.43; g++ (GCC) 16.2.1 20261029 (Red Hat 16.2.1-3); gcovr 8.8.

46.7%: **3162 / 6778** measured branchs covered.

Scope and method

GCC/gcov instrumentation with gcovr, restricted to project src/ files and headers; src/main.cpp, tests and downloaded dependencies are excluded. The executable startup, CLI and real TLS path are additionally exercised by the system validation runner but are not included in this coverage percentage. No exclusions for exception branches or unreachable branches are applied. Compiler-generated C++ exception/cleanup edges remain in the branch denominator. No minimum percentage was specified in the five course PDFs.

Negative-hit parse handling matches the existing CI workaround: --gcov-ignore-parse-errors negative_hits.warn_once_per_file. The collection log is included as coverage-collection.log; inspect warnings when regenerating.

Interactive source coverage · Raw coverage JSON · Summary JSON

File	Covered	Total	Percent
src/api/AdminRoutes.cpp	395	1014	39.0
src/api/Authorization.h	40	86	46.5
src/api/Routes.cpp	998	2352	42.4
src/auth/PasswordHasher.cpp	79	140	56.4
src/auth/PasswordHasher.h	0	0	N/A
src/db/Database.cpp	42	80	52.5
src/db/Schema.cpp	67	138	48.6
src/db/Seed.cpp	64	142	45.1
src/services/AdminTrainService.cpp	373	724	51.5
src/services/AdminTrainService.h	0	0	N/A
src/services/AuditService.cpp	30	66	45.5
src/services/AuditService.h	0	0	N/A
src/services/AuthService.cpp	91	178	51.1
src/services/BookingService.cpp	370	792	46.7
src/services/BookingService.h	0	0	N/A
src/services/TrainService.cpp	263	464	56.7
src/services/TrainService.h	0	0	N/A
src/services/UserService.cpp	119	272	43.8
src/services/UserService.h	0	0	N/A
src/util/Log.cpp	31	58	53.4
src/util/TlsConfig.cpp	21	34	61.8
src/util/TlsConfig.h	3	4	75.0
src/util/Validation.cpp	176	234	75.2

Follow up

Prioritize untested error and rollback paths visible in the HTML report. A passing test suite is not equivalent to complete branch coverage.